

Spotify Tracks Dataset Analysis

Presenters:

Achintya

Shravan Doda

Vrushali Khatane



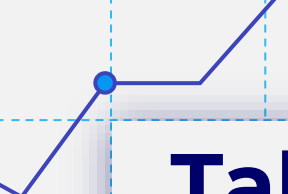


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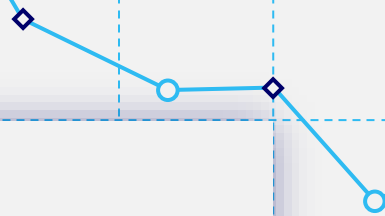
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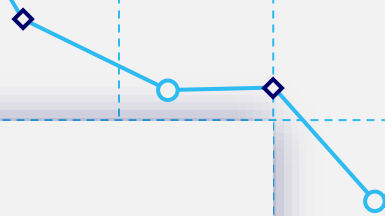


Introduction

- ❑ Platforms like Spotify have revolutionized music access, making it easy for users to discover and enjoy songs anytime, anywhere.
- ❑ The vast amount of data generated by Spotify offers valuable opportunities to uncover patterns in music consumption and understand what makes tracks popular.
- ❑ The project uses a pre-processed subset of the Spotify Tracks dataset to explore key factors influencing track popularity.

The analysis focuses on six key features:

- **Popularity** (target variable).
- **Valence** (happiness level of the track).
- **Loudness** (perceived intensity).
- **Acousticness** (degree of acoustic sounds).
- **Explicit Content** (whether a track is marked explicit).
- **Genre** (categorical classification of the track).



Project Scope

Main Objective

- To predict the **popularity** of a Spotify track based on its musical characteristics and metadata.

Specific Goals

1. Descriptive Analysis:

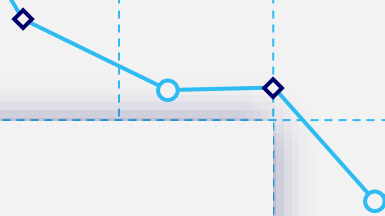
- Summarize and visualize the dataset to understand its structure and key trends.
- Investigate distributions of numerical features (e.g., valence, loudness, acousticness).
- Highlight differences in categorical features (e.g., genre and explicit content).

2. Inferential Analysis:

- Analyze relationships between variables:
 - Example: How does loudness affect popularity?
- Use hypothesis testing to validate insights.

3. Predictive Modeling:

- Build regression models to predict popularity using the selected features.
- Evaluate model performance to identify the most effective approach.

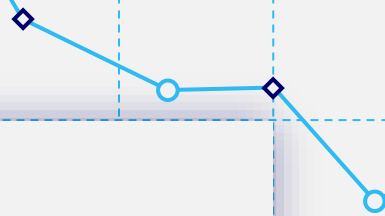


Dataset Description

- Source: Kaggle's Spotify Tracks dataset, containing data for over 90,000 unique tracks across more than 100 genres.
- Purpose: Extract meaningful insights by analyzing a manageable subset of the data.
- Preprocessing:
- Focused on 10 most popular genres to ensure analysis feasibility.
- Selected 6 key features to maintain relevance and simplicity.

Filtered Dataset Stats

- **Track Count:** 7,279 tracks.
- **Explicit Content Distribution:**
 - 6,723 non-explicit tracks.
 - 556 explicit tracks.
- **Genre Distribution:**
 - Top genres include Indian, Jazz, Pop, Hip-hop, etc.
 - Relatively balanced across genres.



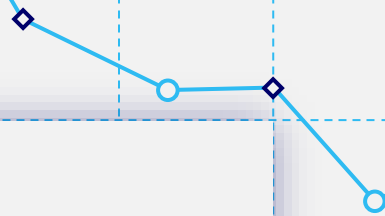
Methods Overview

Descriptive Analysis:

- Summarize numerical features (mean, median, variance).
- Visualize distributions using histograms, KDE plots, and box plots.
- Analyze categorical features using bar charts and contingency tables.

Inferential Analysis:

- Use scatter plots and correlation heatmaps to explore relationships.
- Perform hypothesis tests:
 - **Kruskal-Wallis Test:** Test whether popularity differs across genres.
 - **Mann-Whitney Test:** Compare popularity of explicit vs. non-explicit tracks.
 - **Chi-Square Test:** Examine the dependency between genre and explicitness.



Methods Overview

Predictive Modeling:

- Implement regression models:
 - Multiple Linear Regression.
 - Lasso Regression.
 - Support Vector Regression (SVR).
- Evaluate models using:
 - **R^2 (Coefficient of Determination)**: Measures goodness of fit.
 - **Mean Squared Error (MSE)**: Evaluates prediction error.

Descriptive Analysis

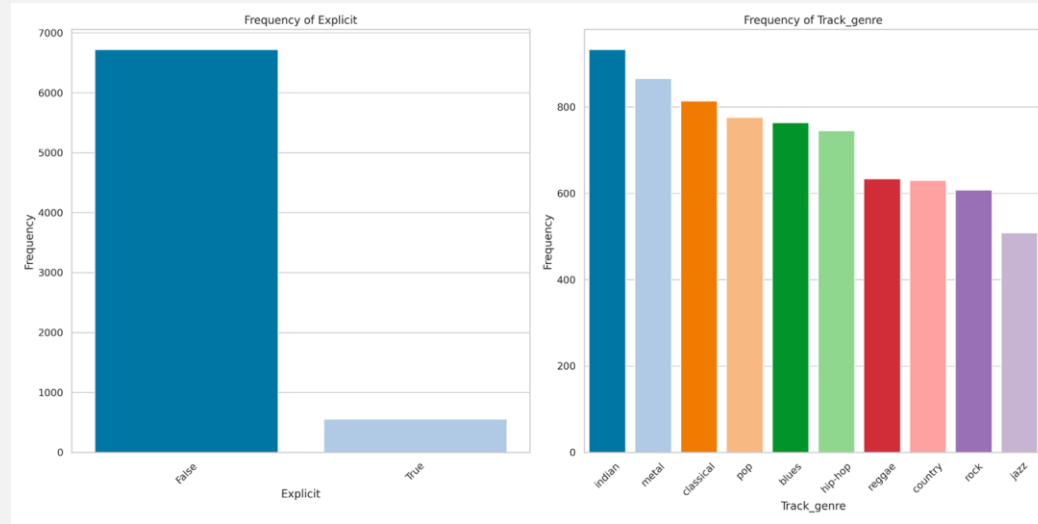
- Valence and Acousticness are bounded within the range [0, 1], with mean values of 0.498 and 0.369, respectively.
- Popularity spans a broader range of [0, 100], with a mean of 26.82.
- Loudness exhibits a mean value of -8.772.

Statistic	Valence	Acousticness	Loudness	Popularity
Count	7279	7279	7279	7279
Mean	0.498	0.369	-8.772	26.820
Std	0.237	0.346	5.733	18.218
Min	0.000	0.000	-41.531	0.000
25%	0.315	0.045	-10.260	16.000
50%	0.497	0.255	-7.102	23.000
75%	0.683	0.685	-5.105	31.000
Max	1.000	1.000	1.023	100.000

Table 3.1: Summary Statistics for Valence, Acousticness, Loudness, and Popularity

Categorical Features

- Categorical Features summarizes the distribution of tracks across different genres.
- The distribution appears relatively balanced, with no single genre dominating the dataset.
- The indian genre contains the highest number of tracks, while the jazz genre has the fewest.
- The dataset includes 6, 723 non-explicit tracks compared to only 553 explicit tracks.



Visualizing Distributions

1. Popularity:

- **Observation:** Right-skewed with most tracks scoring between 20 and 30.
- Very few tracks are highly popular (above 80).

2. Valence:

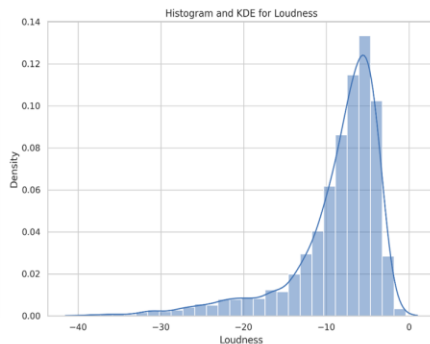
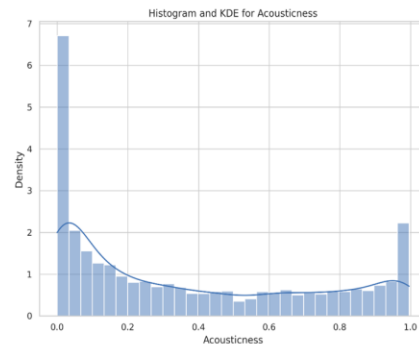
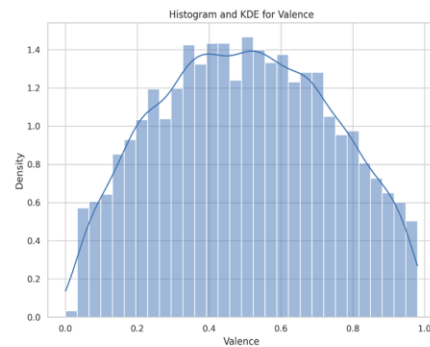
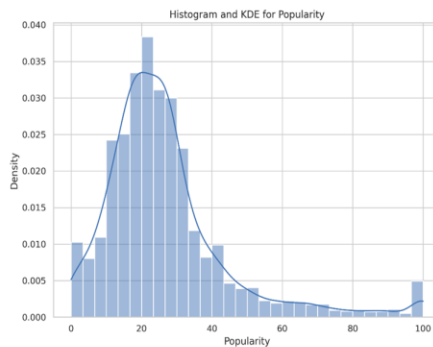
- **Observation:** Uniformly distributed across the range.
- Indicates that tracks span the full spectrum of "happiness" evenly.

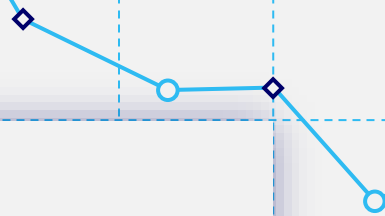
3. Loudness:

- **Observation:** Left-skewed with peaks around -8 to -10 dB.
- Outliers exist at very low loudness levels (below -40 dB).

4. Acousticness:

- **Observation:** Concentrated near zero, showing most tracks are less acoustic.
- Small proportion of tracks with high acoustic values (close to 1).





Outliers and Skewness

Outliers

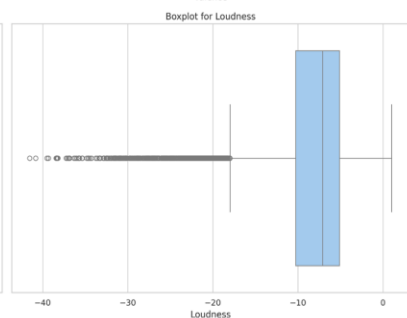
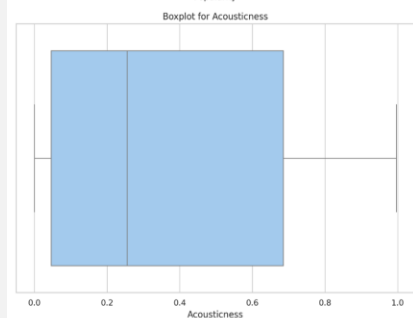
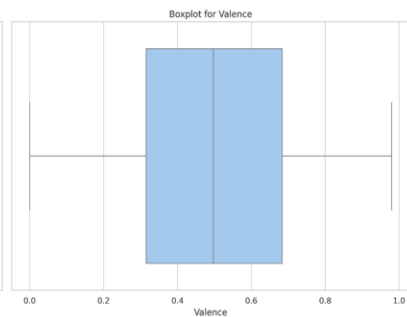
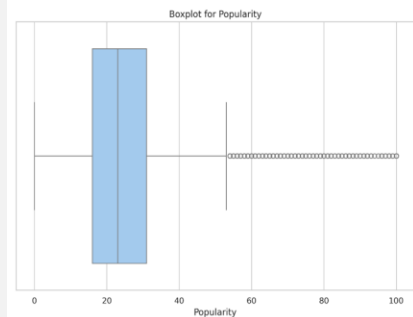
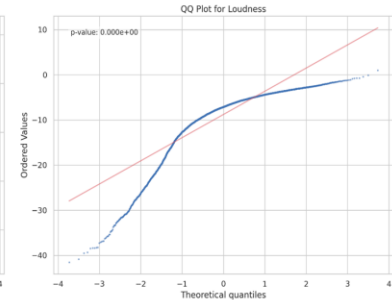
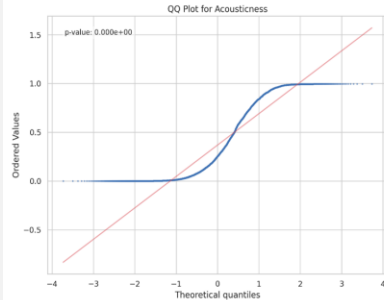
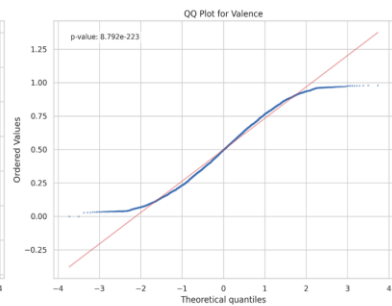
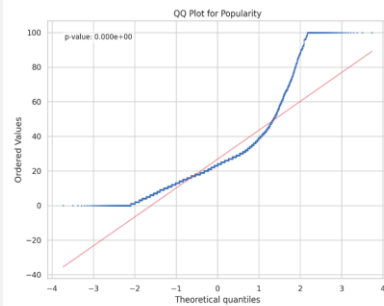
- **Popularity:** High-end outliers (scores near 100), representing disproportionately popular tracks.
- **Loudness:** Low-end outliers (below -40 dB), indicating extremely quiet tracks.
- **Acousticness & Valence:** No significant outliers.

Skewness

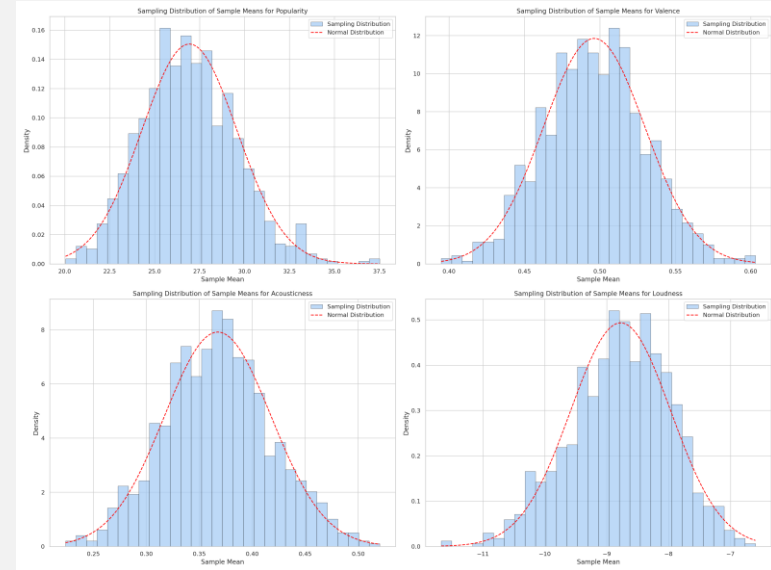
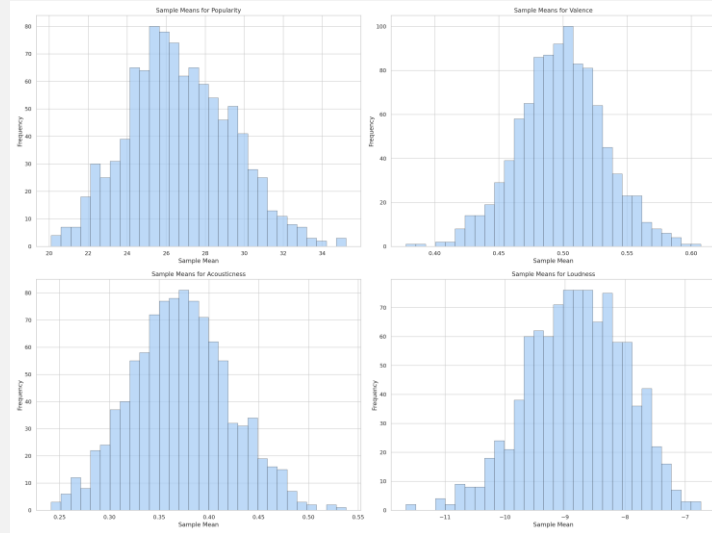
- **Right-Skewed:** Popularity with long tail at high values, clustered around ~26.8.
- **Left-Skewed:** Loudness shifted to low dB; Acousticness peaks near zero.

Impact on Analysis

- **Outliers:** Distort regression coefficients, risk overfitting.
- **Skewness:** Challenges normality assumptions in parametric tests.



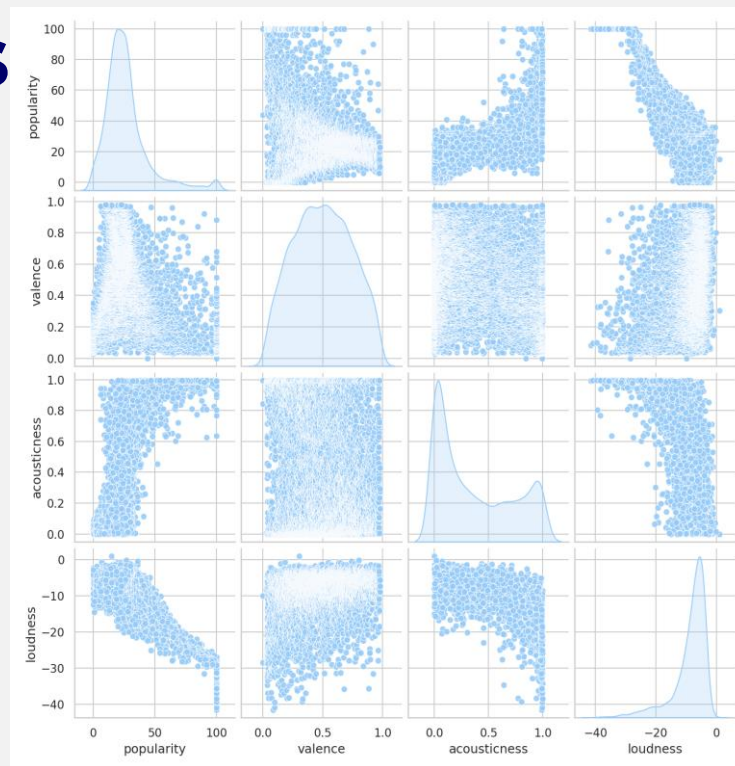
Sampling Distribution and CLT



Inferential Analysis

Scatter Plot Insights

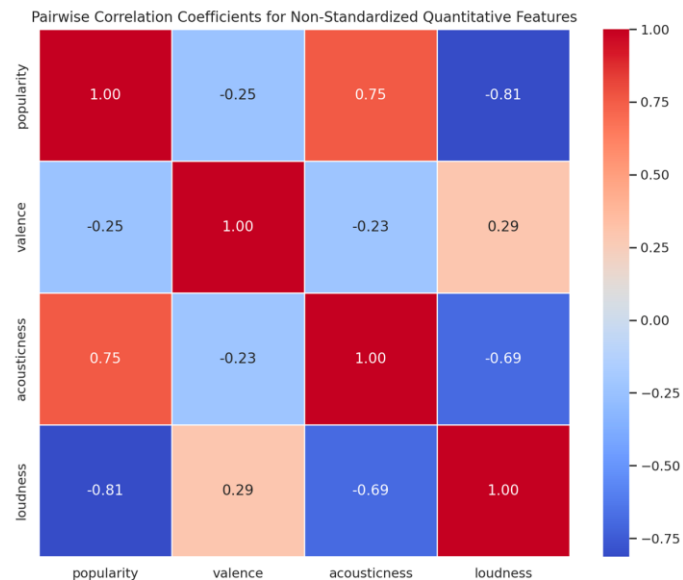
- **Popularity** ↔ **Acousticness**: Positive correlation; more acoustic tracks tend to be more popular.
- **Popularity** ↔ **Loudness**: Strong negative correlation; quieter tracks are more popular.
- **Popularity** ↔ **Valence**: No significant correlation; scattered points with no trend.



Inferential Analysis

Correlation Heatmap Insights

- **Acousticness & Popularity:** Correlation coefficient = 0.75 (strong positive).
- **Loudness & Popularity:** Correlation coefficient = -0.81 (strong negative).
- **Valence & Popularity:** Weak or negligible correlation.



Hypothesis Testing

Test the relationships between **popularity**, **genre**, and **explicitness** using statistical methods.

Tests & Results

1. Genres:

- **Test:** Kruskal-Wallis Test.
- **Result:** *the calculated H-statistic is 4245.986, and the p-value is 0.0.*
- **Implication:** *indicates that Popularity varies with Genre.*

2. Explicit Content:

- **Test:** Mann-Whitney Test.
- **Result:** *the calculated R-statistic is 2980323.5, and the p-value is 0.0.*
- **Implication:** *indicates that Popularity depends on whether the song is explicit.*

3. Genre & Explicitness Dependency:

- **Test:** Chi-Square Test.
- **Result:** *The degree of freedom $df = 9$ and the $\chi^2 = 606.277$ and the p-value comes out to be effectively 0.*
- **Implication:** *indicates that Explicitness and Track Genre may be dependent as well*

Regression Analysis

Predict track popularity using numerical and categorical features.

Multiple Linear Regression (MLR) Results

- **Significant Predictors:**
 - **Acousticness:** Strong positive effect on popularity (+6.845 per unit increase).
 - **Loudness:** Strong negative effect on popularity (-10.603 per unit increase).
 - **Explicitness:** Negative impact (-4.983 for explicit tracks).
- **Insights:**
 - Acoustic, quieter, and non-explicit tracks are more popular.
 - Valence has a negligible effect (non-significant p-value).

Model Performance:

- $R^2 = 0.768$, Mean Squared Error (MSE) = 82.83.

Variable	Coefficient	Std. Error	p-value
intercept	23.992	0.235	0
valence	0.005	0.125	0.681
acousticness	6.845	0.166	0
explicit	-4.983	0.458	3.234e-27
track_genre	0.747	0.046	1.577e-57
loudness	-10.603	0.171	0

Table 4.1: Multiple Regression Coefficients

Lasso Regression

Key Features:

- Shrinks coefficients of less relevant predictors to zero.
- Helps improve model simplicity without compromising interpretability

Results

- **Significant Predictors:**
 - **Acousticness:** +6.835.
 - **Loudness:** -10.509.
 - **Explicitness:** -3.585.
 - **Valence:** Eliminated (coefficient = 0).
- **Insights:**
 - Valence excluded as irrelevant.
 - Predictors have slightly smaller coefficients compared to Multiple Linear Regression due to regularization.

Model Performance:

- $R^2 = 0.767$, $MSE = 83.24$ (similar to MLR).

Variable	Coefficient
const	24.041
valence	0
acousticness	6.835
explicit	-3.585
track_genre	0.708
loudness	-10.509

Table 4.2: Lasso Regression Coefficients

Support Vector Regression (SVR)

- Captures complex relationships.
- Less prone to overfitting compared to traditional linear models.

Results

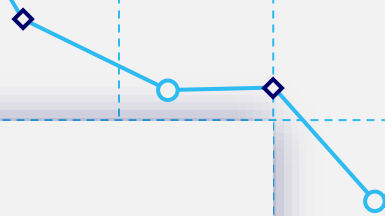
- **Significant Predictors:**
 - **Acousticness:** +7.965 (highest impact).
 - **Loudness:** -7.725.
 - **Explicitness:** -4.845.
 - **Valence:** Moderate positive impact (+0.676).
- **Insights:**
 - Acousticness shows stronger influence in SVR compared to other models.
 - Explicit tracks remain less popular, consistent across all models.

Model Performance:

- $R^2 = 0.742$, $MSE = 92.33$ (less accurate compared to MLR and Lasso).

Variable	Coefficient
const	23.710
valence	0.676
acousticness	7.965
explicit	-4.845
track_genre	0.687
loudness	-7.724

Table 4.3: SVR Coefficients



Summary & Conclusion

Key Insights:

- 1. Popularity Drivers:**
 - Non-explicit tracks with higher acousticness and lower loudness are more popular.
- 2. Significance of Features:**
 - Genre and explicitness are critical categorical predictors.
 - Acousticness and loudness are the most impactful numerical features.
- 3. Model Performance:**
 - **Multiple Linear Regression** and **Lasso Regression** performed best with similar accuracy ($R^2 \sim 0.768$).
 - **Support Vector Regression** captured additional complexities but had lower performance ($R^2 = 0.742$).

Implications:

- Insights can guide artists, producers, and Spotify in optimizing track attributes and playlist curation



Thank You!